Bihar Engineering University, Patna

B.Tech 8th Semester Examination, 2026

Course: B.Tech | *Time: 03 Hours*
Code: 105819 | *Subject: Ad-hoc and Sensor Networks* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which of the following is not a characteristic of the wireless channel?
(i) Path loss
(ii) Interference
(iii) Fading
(iv) Constant Bandwidth

(b) What does MANET stand for?
(i) Multi-Access Network
(ii) Mobile Application Network
(iii) Master Ad-hoc Node
(iv) Mobile Ad-hoc Network

(c) Which of the following is a major issue with TCP over Ad-hoc wireless networks?
(i) High bandwidth
(ii) Centralized routing
(iii) Low latency
(iv) Frequent route breaks and packet loss

(d) IEEE 802.11 is an example of
(i) Scheduling-based protocol
(ii) Multi-channel protocol only
(iii) Contention-based protocol with reservation
(iv) Contention-based protocol

(e) Which one is not a part of the block diagram of wireless sensor network
(i) Error detection Unit
(ii) Power Unit
(iii) Processing Unit
(iv) Communication Unit

(f) Which MAC protocol combines TDMA and FDMA?
(i) CSMA
(ii) IEEE 802.11
(iii) Pure ALOHA
(iv) Hybrid TDMA/FDMA

(g) TESLA is a protocol used for
(i) Data encryption only
(ii) Key distribution
(iii) Intrusion detection only
(iv) Broadcast authentication

(h) Which of the following is a software-based anti-tamper technique?
(i) Watermarking
(ii) Hardware encryption module
(iii) Physical shielding
(iv) Code obfuscation

(i) Biba model is primarily related to
(i) Increase data size
(ii) Decrease security
(iii) Increase latency
(iv) Reduce energy consumption and communication overhead

(j) IEEE 802.15.4 is primarily used for
(i) Cellular networks
(ii) Satellite Communication
(iii) High data rate WLAN
(iv) Low-rate Wireless Personal Area Networks (LR-WPAN)

© Apna BEU
HIGHLY CONFIDENTIAL DOCUMENT - 146 - Government Engineering College, West Champaran @ 25/05/2026 01:55 PM

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

Q.2	(a)	What is the electromagnetic spectrum? Explain its relevance in wireless communication.	[7]
	(b)	Differentiate between Mobile Ad-hoc Networks and Wireless Sensor Networks.	[7]
Q.3	(a)	Explain the architecture of Mobile Ad-hoc Networks.	[7]
	(b)	What are the major design challenges in Ad-hoc and Sensor Networks?	[7]
Q.4	(a)	Explain the working of IEEE 802.11 MAC protocol.	[7]
	(b)	Compare Contention-based and Contention-free MAC protocols.	[7]
Q.5	(a)	Explain Hybrid routing protocols with suitable examples.	[7]
	(b)	What are the issues in designing routing protocols for Ad-hoc networks?	[7]
Q.6	(a)	Explain why standard TCP performs poorly in Ad-hoc wireless networks.	[7]
	(b)	What is data relaying in WSNs? Explain its importance.	[7]
Q.7	(a)	Describe Hybrid TDMA/FDMA based MAC protocols.	[7]
	(b)	Explain CSMA-based MAC protocols used in sensor networks.	[7]
Q.8	(a)	Explain Key Distribution and Management techniques in WSN.	[7]
	(b)	What is Broadcast Authentication in WSN? Mention any one protocol used for it.	[7]

Q.9 Write short note on *any two* of the following: **[7 x 2 = 14]**

(a) IEEE 802.15.4 standard
(b) TESLA protocol
(c) Security Protocols for Sensor Networks
(d) Proactive and Reactive (On-demand) routing protocols

Captured

HIGHLY CONFIDENTIAL DOCUMENT - 146 - Government Engineering College, West Champaran @ 25/05/2026 01:55 PM

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play